



FETAL PERIOD

(9th WEEK TILL BIRTH)

LEARNING OBJECTIVE



- At the end of lecture student should be able to know,
- What is fetal period?
- Factors affecting fetal period.
- Tissue and organ development week wise.
- Milestone.
- Fetal loss.



FETAL PERIOD

- The fetal period begins at the end of the **10th week of gestation (8th week of development)**.
- From the 8th week until birth (around 38 weeks), the developing organism is called a fetus.





FETAL PERIOD

The fetal period is characterized by **two** processes.

- The first is rapid growth (increase in size and cell number).
- The second is continued tissue and organ differentiation (specialization of cells to perform distinct functions).

FACTORS AFFECT FETAL GROWTH



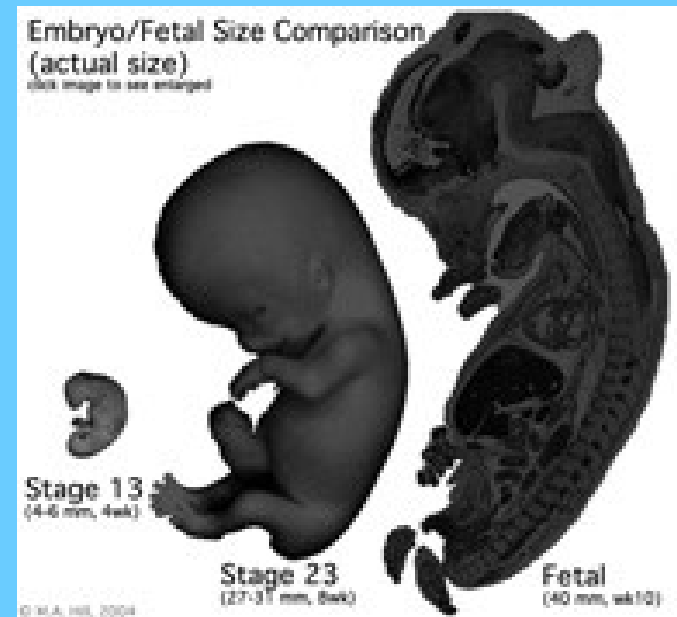
Many factors affect fetal growth,

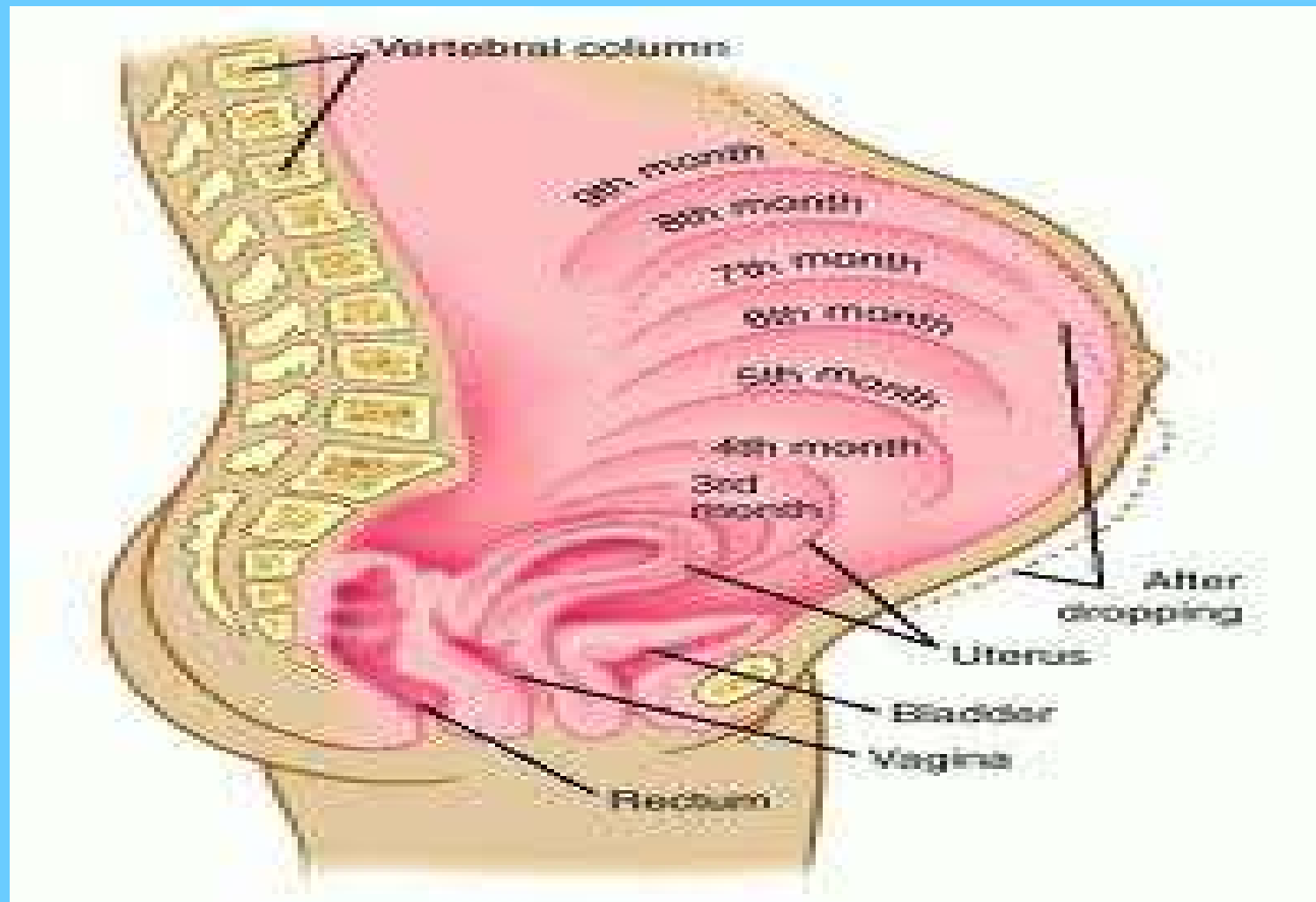
- The nutritional state and social habits (e.g., smoking, drug use) of the mother.
- The state of placental function.
- The genetic makeup of the fetus



- Fetal growth rate is greatest at the beginning of the fetal period (through week sixteen), during which time the fetus increases twenty-five-fold in weight.
- The largest increase in absolute weight gain, however, takes place during the final month of gestation.

RAPID GROWTH



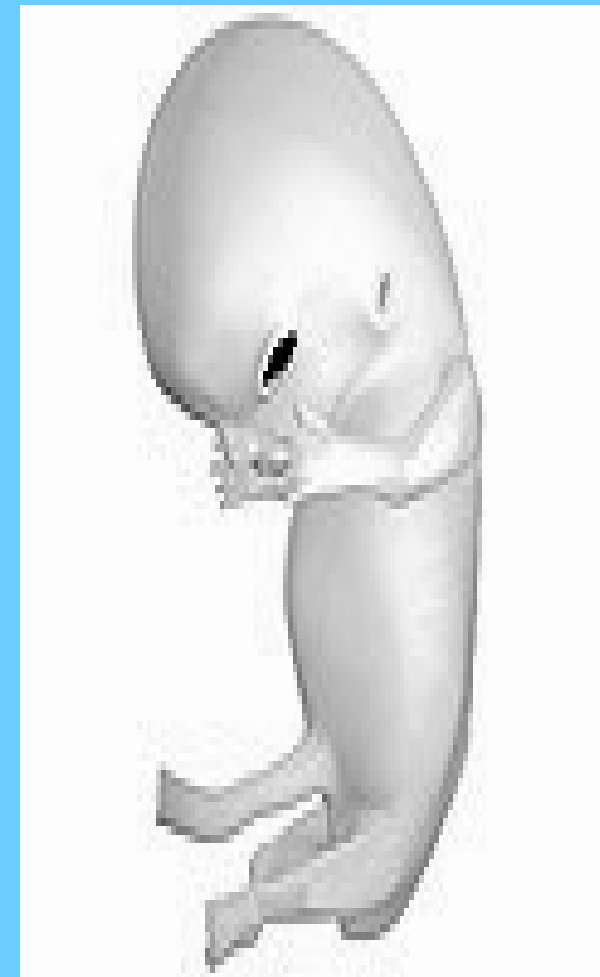




TISSUE AND ORGAN **DEVELOPMENT**

- Gestational age: 9–11 weeks old.
- Embryonic age: Weeks number 8–10. 7–9 weeks old.
- Embryo measures 30–80 mm (1.2–3.2 inches) in length.
- Ventral and dorsal pancreatic buds fuse during the 8th week
- Intestines rotate.
- Facial features continue to develop.
- The eyelids are more developed.
- The external features of the ear begin to take their final shape.
- The head comprises nearly half of the fetus' size.
- The face is well formed
- The eyelids close and will not reopen until about the 28th week.
- Tooth buds, which will form the baby teeth, appear.
- The limbs are long and thin.
- The fetus can make a fist with its fingers.
- Genitals appear well differentiated.
- Red blood cells are produced in the liver.

WEEKS 10–12



Weeks 13 to 16

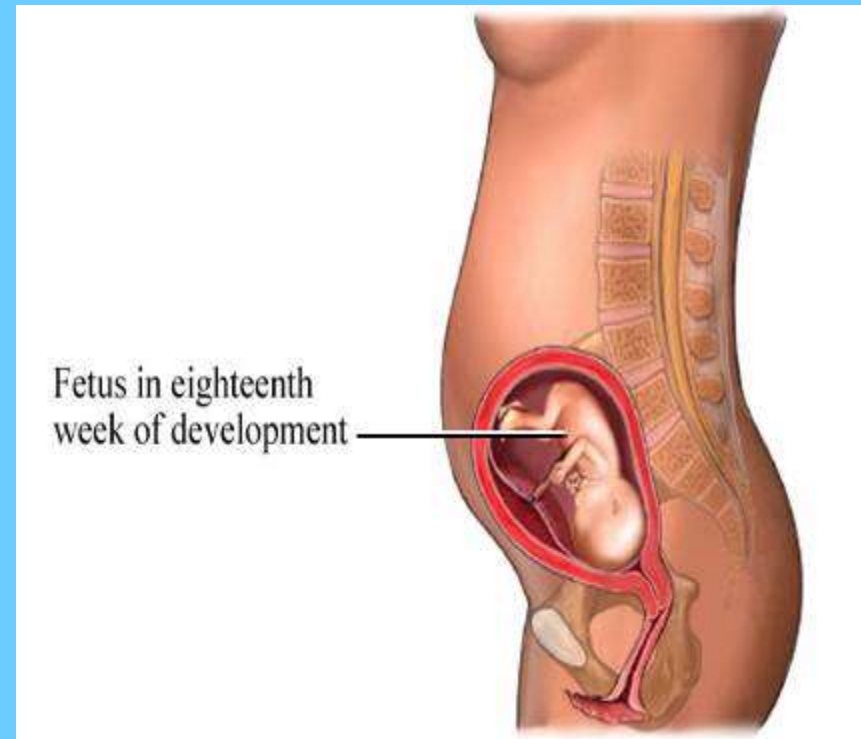


- Gestational age: 12–15 weeks old.
- Embryonic age: Weeks number 11–14. 10–13 weeks old.
- The fetus reaches a length of about 15 cm (6 inches).
- A fine hair called lanugo develops on the head.
- Fetal skin is almost transparent.
- More muscle tissue and bones have developed, and the bones become harder.
- The fetus makes active movements.
- Sucking motions are made with the mouth.
- Meconium is made in the intestinal tract.
- The liver and pancreas produce fluid secretions.
- From week 13, sex prediction by obstetric ultrasonography is almost 100% accurate.
- At week 15, main development of external genitalia is finished



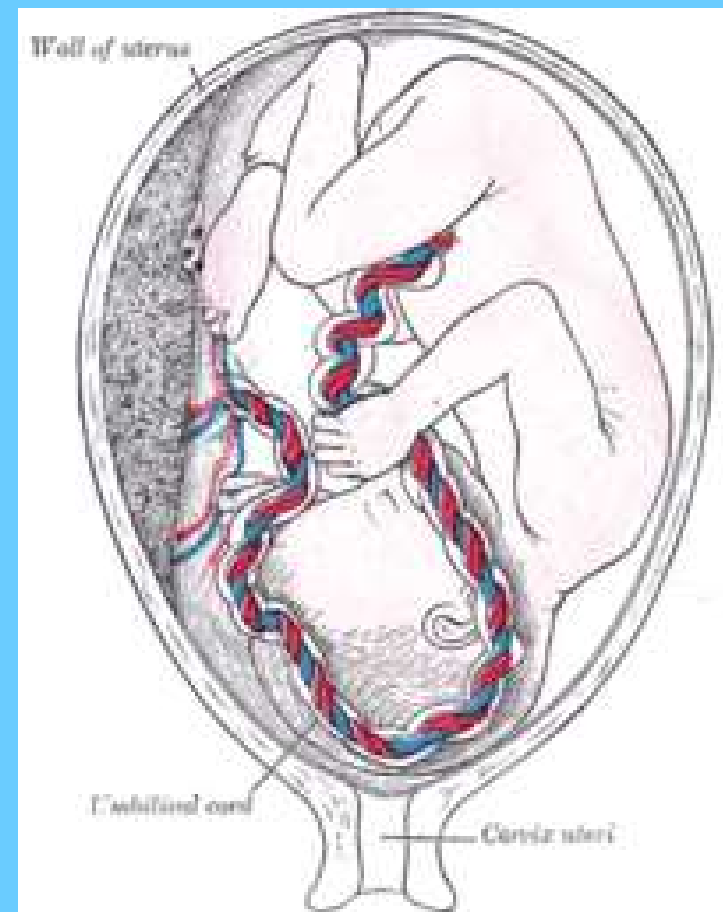
- Gestational age: 18 weeks old.
- Embryonic age: Week numbers 17. 16 weeks old.
- The fetus reaches a length of 20 cm (8 inches).
- Lanugo covers the entire body.
- Eyebrows and eyelashes appear.
- Nails appear on fingers and toes.
- The fetus is more active with increased muscle development.
- "Quickening" usually occurs (the mother and others can feel the fetus moving).
- The fetal heartbeat can be heard with a stethoscope.

WEEK 19



- Gestational age: 22 weeks old.
- Embryonic age: Week numbers 21. 20 weeks old.
- The fetus reaches a length of 28 cm (11.2 inches).
- The fetus weighs about 725 g (1 lb 10 oz).
- Eyebrows and eyelashes are well formed.
- All of the eye components are developed.
- The fetus has a hand and startle reflex.
- Footprints and fingerprints continue forming.
- Alveoli (air sacs) are forming in lungs.

WEEK 23



WEEK 27



- Gestational age: 26 weeks old.
- Embryonic age: Week numbers 25. 24 weeks old.
- The fetus reaches a length of 38 cm (15 inches).
- The fetus weighs about 1.2 kg (2 lb 11 oz).
- The brain develops rapidly.
- The nervous system develops enough to control some body functions.
- The eyelids open and close.
- The cochleae are now developed, though the myelin sheaths in neural portion of the auditory system will continue to develop until 18 months after birth.
- The respiratory system, while immature, has developed to the point where gas exchange is possible.

WEEK 31



- Gestational age: 30 weeks old.
- Embryonic age: Week numbers 29. 28 weeks old.
- The fetus reaches a length of about 38–43 cm (15–17 inches).
- The fetus weighs about 2 kg (3 lb 0 oz).
- The amount of body fat rapidly increases.
- Rhythmic breathing movements occur, but lungs are not fully mature.
- Thalamic brain connections, which mediate sensory input, form.
- Bones are fully developed, but are still soft and pliable.
- The fetus begins storing iron, calcium, and phosphorus.



WEEK 35

- Gestational age: 34 weeks old.
- Embryonic age: Week numbers 33. 32 weeks old.
- The fetus reaches a length of about 40–48 cm (16–19 inches).
- The fetus weighs about 2.5 to 3 kg (5 lb 12 oz to 6 lb 12 oz).
- Lanugo begins to disappear.
- Body fat increases.
- Fingernails reach the end of the fingertips.
- A baby born at 36 weeks has a high chance of survival, but may require medical interventions.





WEEKS 36 to 39

- Gestational age: 35–38 weeks old.
- Embryonic age: Weeks numbers 34–37. 33–36 weeks old.
- The fetus is considered full-term at the end of the 37th week of gestational age.
- It may be 48 to 53 cm (19 to 21 inches) in length.
- The lanugo is gone except on the upper arms and shoulders.
- Fingernails extend beyond fingertips.
- Small breast buds are present on both sexes.
- Head hair is now coarse and thickest.



BIRTH

The process of birth, or parturition (labor), occurs in three stages.

- The first is the **dilation** of the cervix.
- The second is the actual delivery of the fetus.
- The third ends with the expulsion of the placenta.

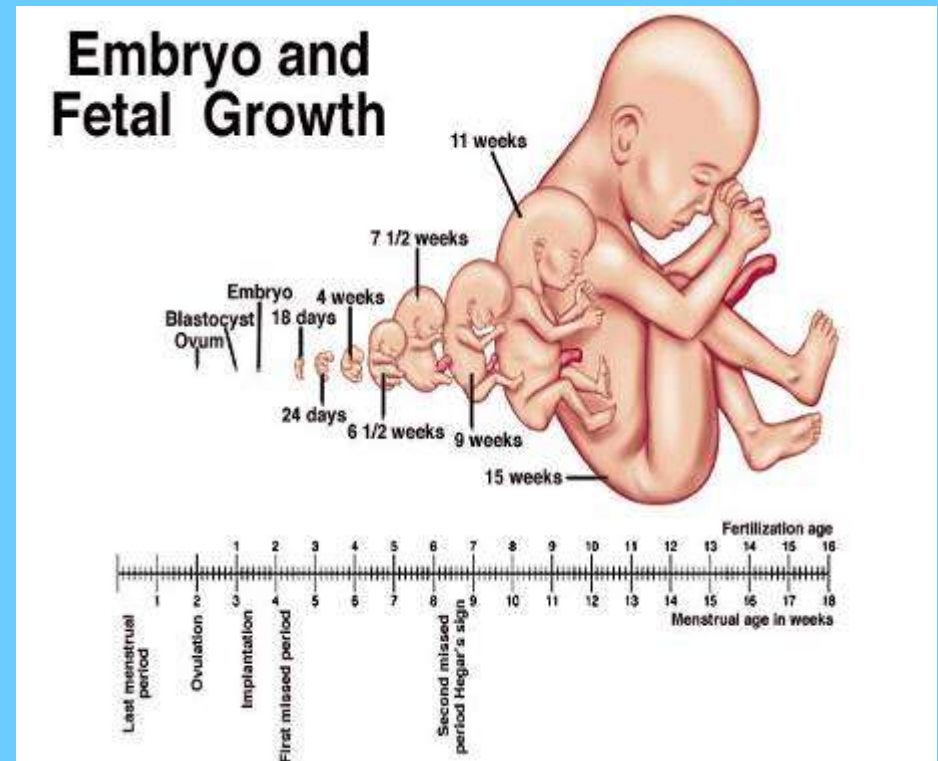
The entire process may take from only a few hours to well over a day to complete.



- The fetus is not as sensitive to damage from environmental exposures as the embryo, and toxic exposures often cause physiological abnormalities or minor congenital malformation.
- All major structures are already formed in the fetus, but they continue to grow and develop.

- The fetus grows dramatically in length and weight during the fetal period.
- The rest of the body grows more rapidly than the head.
- So the relative size of the head shrinks during the fetal period from roughly $\frac{1}{2}$ to $\frac{1}{4}$ of the crown-heel length.

FETAL PERIOD GROWTH

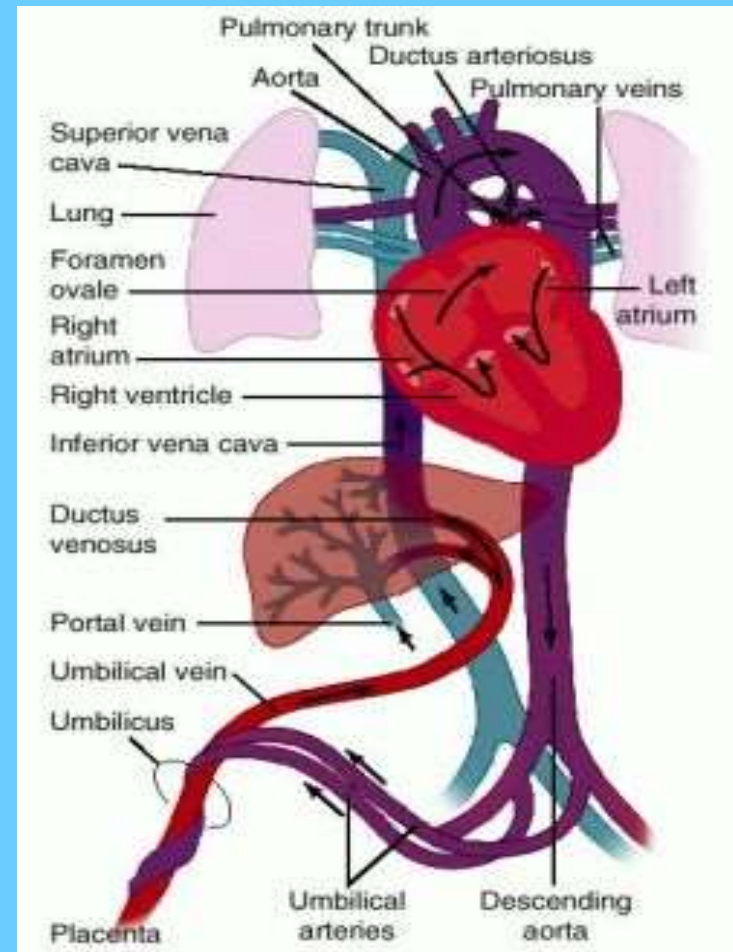




THE MILESTONES OF THE **FETAL PERIOD**

- sounds can be detected as early as 8 or 9 weeks post fertilization (p.f.) with a fetal Doppler ultrasound stethoscope, and 20 weeks post fertilization with a stethoscope or fetoscope.
- An external electronic fetal monitor can be used to obtain fetal heart rate in the third trimester in antepartum testing and intrapartum monitoring.

HEART



OSSIFICATION

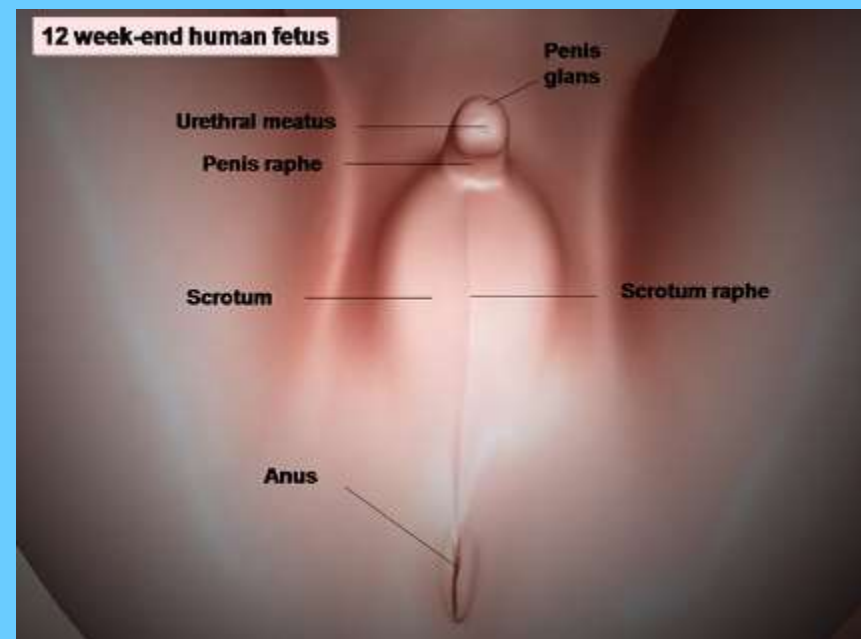
- begins at the end of the embryonic period. Most primary ossification centers of skull, long bones and vertebrae appear by 12 weeks post fertilization.
- The epiphyses of long bones are still cartilaginous at birth.



EXTERNAL GENITALIA

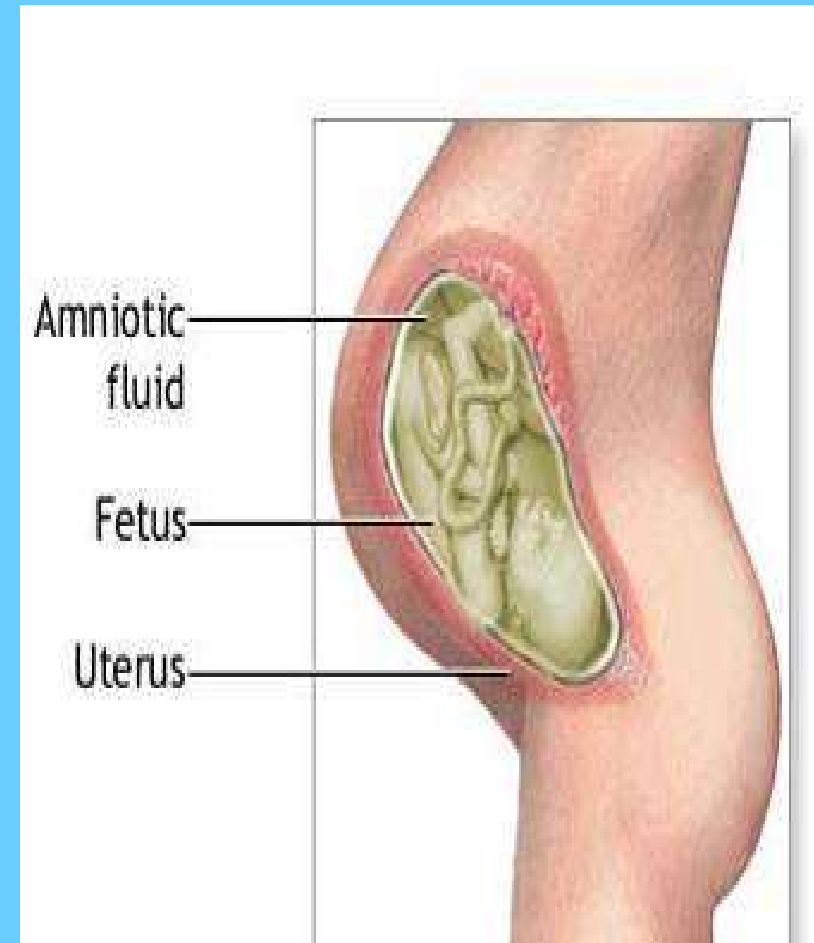


- The external genitalia sexually differentiate during weeks 9-12 post fertilization.



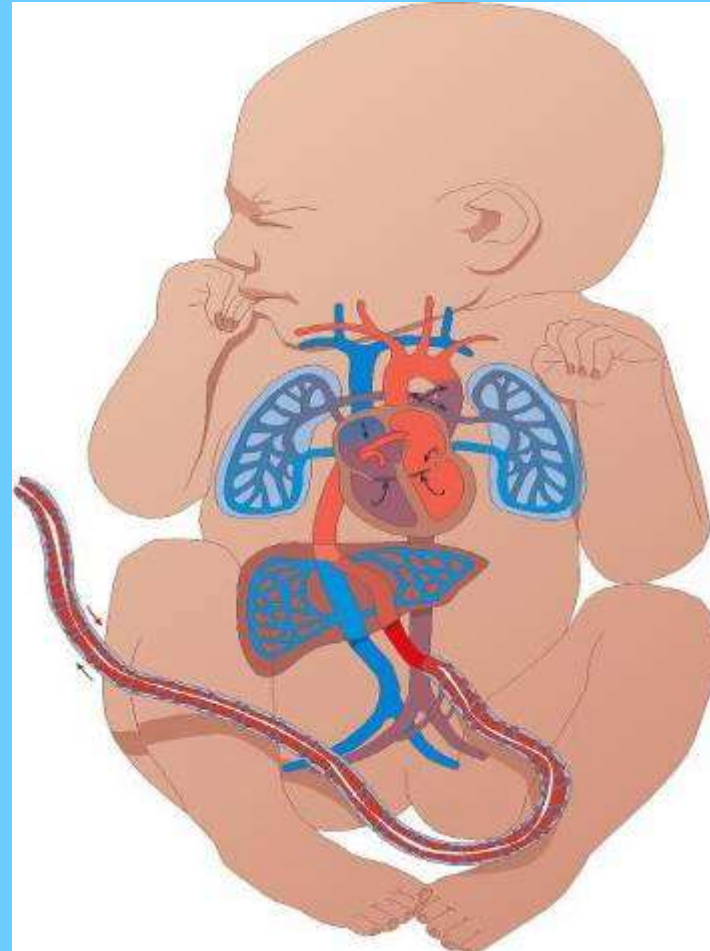
FETAL MOVEMENTS

- Mother first feels fetal movements at about 16-20 weeks post fertilization.
- This is the phenomenon known as "quickening".



- become capable of breathing air at the beginning of the 3rd trimester (about 25-26 weeks p.f.).
- when the lungs enter the saccular phase of development and begin to produce surfactant.
- Lung maturity is the primary determinant of viability.

LUNGS



HEMATOPOIESIS



- begins in the bone marrow in the second trimester.
- In the third trimester bone marrow replaces the liver and spleen as the primary site of hematopoiesis.



EMBRYONIC OR FETAL LOSS

- Embryonic or fetal loss before viability is termed a miscarriage or spontaneous abortion.
- Spontaneous abortion is very common (greater than 50% of pregnancies), especially in the first four weeks post-fertilization.
- Fetal loss after viability is termed stillbirth. Stillbirths account for 0.5-1% of births.



REFERENCES

- Internet different sources